The CMAGIC Cookbook

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Contents

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Color–MAGnitude Intercept Calibration: Type Ia supernova distances from the magnitude at a fixed color.

Contents — Part I (low redshift): §1 the hypothesis; §2 the linear region; §3 modes L/Q/R; §4 K-corrections; §5 extinction; §6 standardization; §7 failure gates and provenance. Part II (high redshift): §8 cross-filter template synthesis; §8.1 the iterative template fit and the insensitivity of B_BV0.6; §9 Mode S (sparse, full covariance); §10 validation (SDSS-II) and application scope; §10.1 the SDSS-II worked example (numbers).

This document describes the method implemented by this package, in enough detail to use it critically. It is based on the two founding papers — Wang, Goldhaber, Aldering & Perlmutter (2003, ApJ 590, 944; hereafter W03) and Wang, Strovink, et al. (2006, ApJ 641, 50; hereafter WS06) — together with a reformulation (three measurement modes and explicit failure gates) developed for archival applications. Users who only want a distance can read §1 and then the README; the pipeline makes every choice described here automatically and reports what it chose.

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1. The physical hypothesis

After maximum light, a Type Ia supernova's B-band magnitude and its B−V color evolve *together*: on the color–magnitude diagram (B versus B−V), the supernova travels a remarkably straight line for much of the first month past maximum, as the ejecta expand and cool. The CMAGIC hypothesis is:

> The B magnitude at a fixed color, B−V = 0.6 mag — written B_BV0.6 — is a calibratable standard candle, in exactly the sense that the peak magnitude is the standard candle of light-curve fitters such as SALT3.

Why measure brightness at fixed *color* rather than at fixed *time* (peak)?

1. The locus is straight and slowly traversed. The linear color–magnitude relation (slope β_BV = 1.94 ± 0.16 after K-corrections; W03) is measured from many nights of data, not from the single moment of peak, so B_BV0.6 is statistically robust and does not require dense coverage of maximum light.
2. Reduced extinction sensitivity. Dust moves a supernova *along* a vector of slope R_B ≈ 4.1 in the same diagram. The extinction correction to B_BV0.6 is therefore (R_B − β_BV)·E(B−V) ≈ 2.2·E rather than R_B·E ≈ 4.1·E — roughly half the sensitivity of a peak-magnitude method to reddening errors (W03).
3. Smaller intrinsic dispersion. In the low-extinction calibration sample, B_BV0.6 standardizes to σ ≈ 0.07–0.10 mag, versus ≈ 0.10 mag for B_max (W03, Table 2).
4. Independence. The residuals of CMAGIC and peak-magnitude distances correlate only weakly (Pearson ≈ 0.15; WS06), so the two are usefully independent distance estimators from the same photometry.

The distance chain is then conceptually identical to any standard-candle method:

```
B_BV0.6  →  K-correction  →  extinction correction  →  M_BV(Δm15) calibration  →  μ, D
```

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2. The color–magnitude diagram and the linear region

Plot nightly B against nightly B−V. The locus has four parts:

- a pre-maximum branch (distinct; hysteresis relative to the post-max branch),
- a post-maximum shoulder in the first days after B maximum (curved; W03 exclude a two-day gap around the transition),
- the linear region — the CMAGIC branch proper, typically spanning B−V ≈ 0.2–1.0 for an unreddened object,
- the color turnback, when B−V reaches its maximum (t ≈ +30 d for normal decliners) and turns blueward; beyond it the relation is no longer the same line.

The fiducial window is decline-rate-scaled (WS06):

subtype	window (days past B max)
normal	[$7/\Delta m_{15}$, $30/\Delta m_{15}$]
91T-like	[$10/\Delta m_{15}$, $29/\Delta m_{15}$]

For Δm_{15} = 1.1 this is +6.4…+27.3 d. The package intersects this window with an object-specific safety exit at t(B−V max) − 3 d, computed from the object's own smoothed color curve, which protects objects with unusually early or late color maxima. Within the window, nights are combined by weighted averaging; no residual-based point rejection is applied beyond a conservative 3σ photometric clip.

Two practical caveats established on archival data:

- The window inherits Δm_{15} quality. A poorly measured decline rate (e.g., a light-curve fit with the stretch parameter at its bound) produces a nonsense window; the pipeline gates on Δm_{15} provenance.
- Slow decliners. For $\Delta m_{15} \lesssim 0.9$ the $7/\Delta m_{15}$ entry can admit the tail of the post-max shoulder, mildly biasing a free-slope fit. The calibrated chain (fixed slope, §3) is robust to this at the few-percent level in distance; the free slope is reported as a diagnostic, not used in the distance.

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3. The three measurement modes

The goal is always the same number — the magnitude at B−V = 0.6 — but real light curves come in three conditions. The pipeline chooses the mode automatically and records the choice.

Mode L — linear (the classical method)

When the linear region is well populated, fit a weighted straight line to B versus B−V in the window and evaluate it at B−V = 0.6:

```
B(c) = B_BV0.6 + β_BV (c − 0.6)
```

The calibrated chain fixes β_BV = 1.94 (the K-corrected population slope; W03) — this is what the M_BV calibration of §6 was constructed with. The free slope β_free is also fit and reported with its error as a *diagnostic*: values far from 1.94 on clean, single-source photometry indicate either peculiarity or time-variable extinction (see §7's SN 2006X note). A free-slope value is considered valid only if the fit has ≥ 5 nights, color span ≥ 0.25 mag, no internal color gap > 0.4 mag, rms ≤ 0.15 mag, and β_free within the WS06 validity cut [1.5, 2.5]. Per-fit error normalization (v0.3.1, the PI's prescription): the errors of the CMAGIC fitting parameters must be normalized to $\chi^2/\mathrm{dof} = 1$ per fit — after every weighted fit (Modes L, Q, R, and the Mode S GLS with n ≥ 2, where n points give dof = n − 1 for the mean), the fit-parameter errors are scaled by $s = \max(1, \sqrt{(\chi^2/\mathrm{dof})})$; χ^2/dof and s are stored in the result and printed on the diagnostic panel. A single-point Mode S has no dof: it relies on the sample-level term and says so in its flags.

Mode Q — quadratic interpolation

When the linear region is not obvious — sparse windows, curved (parabolic) loci, and 91bg-like objects for which W03 showed the linear model fails — B_BV0.6 is still perfectly well defined *as the magnitude where the locus crosses B−V = 0.6*. Mode Q fits

```
B(c) = a₀ + a₁ (c − 0.6) + a₂ (c − 0.6)²
```

over the post-maximum branch and reports $a_0 \pm \sigma$. Interpolation only: the observed colors must bracket 0.6 (or approach it within 0.05 mag), otherwise the mode fails with a named gate (`no_bracket`). For 91bg-like objects ($\Delta m_{15} \geq 1.7$) the *measurement* is delivered but the *standardization* is flagged `uncalibrated`: the M_BV(Δm_{15}) calibration of §6 does not extend to them.

Mode R — heavily reddened objects

When extinction pushes the entire observed branch redward of B−V = 0.6, extrapolating the fit back to 0.6 is forbidden — extrapolation across a magnitude or more of color is precisely the systematic that corrupts naive fits. Instead (PI prescription):

1. Estimate the reddening from the peak color: E_guess = B_max − V_max (the intrinsic peak color of a normal Ia is ≈ 0; optionally refine by subtracting the intrinsic color locus $\mathcal{E}_0(\Delta m_{15})$ of Eq. W03-8 below).
2. Measure the magnitude at the shifted target color c* = 0.6 + E_guess, by Mode L or Mode Q, strictly by interpolation within the observed branch. Record (m*, c*, E_guess, β_used).
3. When the true reddening E_true is later determined (from §5, from external data, or from a dust-law analysis), apply the deferred correction:

```
B_BV0.6 = m* − β_used (c* − 0.6 − E_true) − 0.6 β_used − R_B E_true + 0.6 β_used
        = m* − β_used (c* − 0.6) + (β_used − R_B) E_true
```

For a linear locus this is algebraically identical to the classical chain (measure at observed 0.6, correct by (R_B − β)E) — Mode R changes *where the locus is evaluated*, never the physics — but it avoids the extrapolation entirely. The host dust law R_B is configurable per object (default 3.1; e.g., SN 2006X requires its known anomalous R_B ≈ 2.5, and Mode R makes the dust law the *only* open parameter for such objects).

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4. K-corrections (the WS06 scheme)

Five raw outputs receive K-corrections: B_max, V_max, Δm_{15}, B_BV0.6, and β_BV. Following WS06 (their Appendix D): each correction is linear in the supernova's raw color E, with an intercept $\mu^k(z)$ and slope $\psi^k(z)$ that are cubic polynomials in redshift, calibrated from a spectrophotometric library of reddened/unreddened light-curve pairs. The package evaluates the polynomials at the object's z and raw color. At z ≤ 0.03 the corrections are ≤ 0.01 mag; they are always applied and always quoted. (Reproduction note: the distance-relevant K(B_BV) reproduces the WS06 worked example to ~1 mmag; the published per-band table entries for B_max/V_max include light-curve-refit effects beyond the tabulated polynomials, an ambiguity ≤ 0.01 mag in μ at these redshifts, documented in the validation suite.)

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5. Extinction

Milky-Way extinction is removed first (Cardelli law, R_B = 4.15, with E(B−V)_MW from the Schlafly & Finkbeiner maps). The host component uses CMAGIC's internal estimator (W03 Eqs. 7–9): the color-excess measure

```
Ecal = (B_max − B_BV0.6) / β_BV                                  (W03 Eq. 7)
Ecal₀ = (−0.118 ± 0.013) + (0.249 ± 0.043)(Δm₁₅ − 1.1)           (W03 Eq. 8)
E_host = Ecal − Ecal₀                                               (W03 Eq. 9)
```

with $\mathcal{E}_0$ the intrinsic (dust-free) locus of the same quantity as a function of decline rate. The host correction to the candle is A_BV = (R_B − β_BV)·E_host with the host-law R_B = 3.1 by default (override per object when the dust law is known to be anomalous). E_host absorbs *color calibration errors* of the photometry as if they were dust — the reason the pipeline enforces single-source photometry (§7).

5.1 The two extinction philosophies

Supernova cosmology has always had two ways to handle a red supernova, and the two teams whose samples discovered the accelerating expansion each used one of them. The empirical-extinction route — the original Riess et al. (High-z Team / MLCS) analyses — estimates the extinction of each object from its own photometry as a color excess over an intrinsic locus, then corrects it through a known extinction law (an assumed R_B). The Tripp route (Tripp 1998; the standardization of Perlmutter et al.'s SCP lineage, and of SALT2/SALT3 today) does not ask *why* an object is red: it fits one linear color coefficient β on the whole sample and lets the regression decide the correction.

What each buys, and what each risks:

- The empirical route is physically explicit and per-object, and it preserves whatever signal remains in the residuals. Its systematic is the assumed law: SN 2006X's anomalous dust (R_B ≈ 2.48) corrected with the standard 3.1 produces a 0.82 mag distance error from the law alone (§3, Mode R).
- The Tripp route is self-calibrating and agnostic, but its fitted β is a variance-weighted compromise between two physically different slopes: the dust vector (R_B ≈ 4.1) and the intrinsic color–luminosity slope. One β is correct only if the dust-to-intrinsic mixture of the sample's colors is universal. When selection changes that mixture with redshift — a magnitude-limited survey keeps losing dusty objects toward high z — the correct effective β changes along the redshift axis while the applied one does not, and the mismatch leaks into the Hubble diagram. That is the mechanism behind the SALT3 redshift drift dissected in §10.3, where the sample-fitted β = 2.41 ± 0.20 sits well below both the fiducial 3.1 and the dust value — the signature of a blended color.

CMAGIC's position in this dichotomy is distinctive on both counts:

1. The pipeline carries both routes. The internal W03 estimator above (E_host from $\mathcal{E} - \mathcal{E}_0$, corrected through the known law as (R_B − β_BV)·E_host) is the empirical route, per object. The sample-level fitted β_C·c of §6.1 is the Tripp route. The precedence rule of §6.1 selects which one is in force — never both — so the user always knows which assumption they are buying: a known dust law, or a universal color–luminosity relation. Disagreement between the two routes on a single object is itself a diagnostic; it is what flagged SN 2006X.
2. The dust lever arm is halved by construction. Extinction enters B_BV0.6 as (R_B − β_BV)·E ≈ 2.2·E rather than R_B·E ≈ 4.1·E, so whichever route errs, the damage to a CMAGIC distance is roughly half of what the same error does to a peak-magnitude method. This is also why the fitted color term on CMAGIC residuals is small (β_C = +0.94 ± 0.35 on the SDSS-II sample, §6.1): most of the color sensitivity is removed before standardization ever sees it.

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6. Standardization and the distance

The absolute calibration is the W03 Table 3 linear law M_BV(Δm_{15}) = a + b·Δm_{15}, constructed on color-restricted subsamples (B_max − V_max ≤ 0.05 / 0.2 / 0.5) with dispersions 0.07–0.16 mag, on the H_0 = 65 scale (rescaled explicitly; the H_0 dependence is a single coherent 5 log(H_0/65) shift). The WS06 refinement — a bilinear decline-rate law with a kink at Δm_{15} = 1.1, which beats a quadratic at 43:1 odds — is available as an option (zero point anchored to W03 Table 3 at Δm_{15} = 1.1). Then

```
μ = B_BV0.6(corrected) − M_BV(Δm₁₅),      D = 10^{(μ−25)/5} Mpc.
```

Error budget per object: the fit error on B_BV0.6 (typically 0.02–0.06 mag on good archival photometry), the calibration dispersion (0.07–0.16 mag, dominant), the extinction-estimator error propagated through $(R\_B - \beta)$, and the K-correction ambiguity (≤ 0.01 mag at $z \leq 0.03$).

6.1 Sample-level fitted corrections (v0.3)

The PI's rule, verbatim: "the correct way is to always fit the x1 and c correction so that any correlation is removed." Whenever a SAMPLE of CMAGIC distances is assembled, the standardization of §6 gains fitted shape and color terms:

```
resid_i = M0 + delta_S·[mode_i = S] + alpha_C·x1_i + beta_C·c_i
```

fit by robust weighted least squares against the Hubble residuals (`cmagic.standardize_sample`; covariates from the object's own engine — the SALT3-NIR iterative (x1, c), or the Hsiao stretch mapped through $s \approx 0.98 + 0.091$ x1 with the near-peak color-warp tilt as the c proxy — or supplied externally, e.g. from an independent SALT3 run). The mode-offset term delta_S repairs the L-versus-S zero-point split found in the §10.1 comparison; coefficient covariance is propagated into the corrected errors, and per-mode errors are inflated so the post-fit chi^2/dof is unity per mode. The post-fit weighted residual correlations against x1 and c are zero by construction — that is the acceptance criterion, and the test suite asserts it.

Precedence (no double color correction). When beta_C is fitted, the internal W03 Eqs 7–9 host-extinction correction is backed out of the base magnitudes — the fitted color term empirically absorbs what the internal E_host estimator measures (and, on synthesized high-z peaks, what it misses). E_host remains reported for diagnostics, but its correction applies only when the user excludes 'c' from the covariates. This toggle is exactly the choice between the two extinction philosophies of §5.1: covariates with 'c' = the Tripp route, without = the empirical-extinction route.

Single-object caveat. A single supernova is never corrected by fitted terms: `distance()` output is estimator-pure, and the fitted corrections are a sample-level operation — exactly as SALT3's alpha/beta are training products, not per-object physics.

SDSS-II numbers (v0.3.1 per-fit-normalized errors; 28-object fit set; 6 objects gated by the boundary-hit rule, snids 1794/2017/2031/2440/2635/2992, and 2030 excluded for its bound-hit external covariate): M0 = $+0.145 \pm 0.036$, delta_S = -0.101 ± 0.063, alpha_C = -0.056 ± 0.038, beta_C = $+0.941 \pm 0.346$. Pre → post: rms 0.246 → 0.213 (mode L 0.181 → 0.165, mode S 0.292 → 0.263); chi^2/dof 2.81 → 2.05, and 1.0 per mode after inflation. Post-fit weighted correlations: r(x1) = 0.00, r(c) = 0.00.

The per-fit normalization and the sample factors (honest accounting). The v0.3.1 per-fit rescaling barely moves the sample-level inflation (L 1.284 → 1.270, S 1.634 → 1.624): the per-object distance error is dominated by the calibration dispersion (0.07–0.16) and the host-estimator term (≈ 0.09), so even a $\times 1.5$ rescaling of the fit-parameter error (e.g. snid 5103: χ^2/dof = 2.2, eB_BV0.6 0.049 → 0.073) shifts the total error by only a few thousandths of a magnitude. The residual sample factors therefore measure genuine population/intrinsic scatter plus synthesis systematics — not fit inconsistency, which is now normalized away per fit as prescribed.

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7. Failure semantics and provenance (what makes this pipeline safe to use blind)

Every fit either returns a distance or a named gate, never a silently degraded number:

gate	meaning
`window_empty`	the decline-rate window contains no valid nights (e.g., 91bg-like: the color maximum arrives so early that no linear branch exists — that *is* the physical result)
`too_few_nights`	< 4 nights in the window
`no_bracket`	Mode Q/R interpolation impossible: the branch never crosses the target color
`rms_gate` / `beta_gate`	Mode L quality gates failed
`dm15_provenance`	decline rate unmeasured or from a bound-hit fit: no trustworthy window
`photometry_defect`	upstream data flagged (duplicated columns, cross-instrument artifacts)

Provenance rules learned from archival forensics, enforced automatically:

- One photometric source per fit (never mix observatories in a band; preference order CSP > CfA > KAIT/LOSS > other; Swift/UVOT optical bands are never used). Mixed compilations inflate the CMAGIC rms three- to five-fold and bias the internal extinction estimator.
- De-duplication: rows duplicating the same publication via different aggregators are dropped by bibcode matching.
- Every result carries: source used, nights selected, phase and color spans, mode, β_free ± σ, rms, all gates evaluated, and the diagnostic panel (locus, window boundaries, fit, color-curve inset, annotation block).

A diagnostic worth knowing about: on clean single-source photometry, a *valid* free slope significantly different from 1.94 carries physics. SN 2006X — with documented time-variable circumstellar absorption — shows β_free = 2.25–2.28 (rms at calibration quality), intermediate between the intrinsic slope and its anomalous dust vector R_B ≈ 2.5: time-variable extinction slides points along the dust vector. The free slope is CMAGIC's built-in dust-variability alarm.

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- Conley, A., et al. 2006, ApJ, 644, 1 (the blind CMAGIC cosmology at high z)

- DES Collaboration 2024, ApJ, 973, L14; Sánchez, B. O., et al. 2024, ApJ, 975, 5 (DES-SN5YR release, github.com/des-science/DES-SN5YR)

Part II — CMAGIC at High Redshift

At $z \lesssim 0.1$, single-filter K-corrections suffice and Part I applies as written. At higher redshift the observer-frame filters no longer sample the rest-frame B and V bands at all: cross-filter K-corrections are required — effectively interpolations of the observed multi-band photometry, guided by a spectral template, from the observer frame to the rest-frame B and V light curves that CMAGIC needs. This part formulates that extension, its sparse-data estimator, and its validation. There is direct precedent: Conley et al. (2006, ApJ 644, 1) measured (Ω_m, Ω_Λ) from a blind CMAGIC analysis of 21 high-z SNe — on "data sets not observed in a manner optimized for CMAGIC" — and confirmed the acceleration through the color–magnitude channel.

8. The cross-filter problem and the template-synthesis solution

For a supernova at redshift z observed in filters $\{X_1 \ldots X\_n\}$ (Rubin/LSST ugrizy, Roman WFI, JWST NIRCam, or the standard UBVRIZ/ugriz/JHK sets), rest-frame B at rest phase p samples the SED near 4400(1+z) Å in the observer frame — between, or beyond, the observed bands. The classical per-band cross-filter K-correction (Kim, Goobar & Perlmutter 1996) transfers each observed magnitude to a chosen rest band. We adopt the equivalent but more robust template-synthesis formulation, which the pipeline implements end to end:

1. Rest-frame clock. Phases are computed as $p = (t - t_0)/(1+z)$: cosmological time dilation is applied before anything else, and the CMAGIC window of §2, the decline rate Δm_{15}, and the color-curve turnback are all defined in *rest-frame* days.
2. Template. A spectral time series F_T(p, λ) of a normal Ia — either the Hsiao template (the simple default the method needs nothing fancier than) or the SALT3-NIR model (Pierel et al. 2022), whose wavelength coverage extends into the near-infrared and is preferred when the observed bands redshift beyond ~8500 Å rest (Roman and JWST at $z \gtrsim 1$, where rest-frame B–V is sampled by observer-frame YJH).
3. Warp to the data. The template is redshifted — F_T(p, λ(1+z))/(1+z), i.e. wavelengths stretched and the SED dimmed per bandpass convention — and constrained by the observed photometry: in Hsiao mode, a stretch (from the fitted rest-frame decline) plus a smooth per-epoch color warp (a grey term and an extinction-law-shaped tilt) fitted so the template's synthetic magnitudes reproduce every observed band at every epoch; in SALT3-NIR mode, the standard (t_0, x_0, x_1, c) fit. The warp is smooth in wavelength by construction, so it interpolates *between* the observed bands without inventing spectral features.
4. Synthesize rest-frame B, V. At each observation epoch, rest-frame B and V magnitudes are synthesized from the warped template through the Bessell B and V transmissions. The output is exactly what Part I consumes: nightly B and B−V, now in the rest frame, with the K-correction, the (1+z) dilation, and the wavelength stretch all inside the synthesis.
5. Errors and covariances by Monte Carlo. The observed photometry is perturbed by its errors (default 200 realizations), the warp and synthesis are redone per realization, and the empirical covariance of the synthesized (B_i, V_i) — including the B–V covariance within an epoch and correlations between epochs induced by the shared warp — is propagated into the CMAGIC fit. A

template-choice systematic (Hsiao versus SALT3-NIR difference, in the bands actually used) is reported alongside.

Filter transmissions: the package ships with (or registers on demand from `sncosmo`) the Bessell UBVRI, SDSS/LSST ugrizy, Roman WFI, and 2MASS JHK curves; any other filter — JWST NIRCam included — is supplied as a two-column (wavelength, transmission) file via `cmagic.highz.register_filter()`. The inputs per supernova are then exactly as the design requires: the light curves, the filter transmissions, and the redshift.

9. Sparse light curves: the fixed-slope estimator with full covariance

High-z light curves are often sparse: one or two epochs may land in the CMAGIC linear window. The linear fit of Mode L is then impossible — but the hypothesis of §1 does not require fitting the slope, because the slope is a population constant. Mode S (sparse): with β fixed at 1.94, every single point in the window is an estimate of the candle,

```
m_i = B_i − β (c_i − 0.6),   c_i ≡ B_i − V_i
    = (1−β) B_i + β V_i + 0.6 β
```

with variance following from the epoch's (B, V) covariance matrix — note that B appears on both axes of the diagram, so the B–V covariance term is *not* optional:

```
Var(m_i) = (1−β)² Var(B_i) + β² Var(V_i) + 2β(1−β) Cov(B_i, V_i)
```

Multiple sparse points are combined by generalized least squares with the full covariance

```
V_ij = Cov(m_i, m_j)  +  σ_β² (c_i − 0.6)(c_j − 0.6)  +  δ_ij σ_int²
```

where the first term carries the synthesis-induced inter-epoch correlations (§8.5), the second propagates the population slope uncertainty $\sigma_\beta = 0.16$ — *fully correlated* across the epochs of one supernova, and vanishing for points at the fiducial color — and σ_{int} is the intrinsic scatter of the locus about the line (set from the low-z training sample). The slope term makes explicit what sparse data cost: a single point at $c = 0.9$ pays $0.16 \times 0.3 \approx 0.05$ mag of slope systematic; a point near $c = 0.6$ pays almost nothing — so the *placement* of the surviving epochs, not just their number, sets the error. Mode S reports n_window, the color leverage $|\bar{c} - 0.6|$, and the slope-systematic share of the error budget, and it is gated exactly like the other modes (`window_empty` when nothing survives; no silent degradation).

10. Validation and application scope

Validation: the SDSS-II supernova survey (ugriz light curves, $z \approx 0.05$–0.4, published photometry and redshifts) — deep enough that cross-filter synthesis is genuinely exercised, low enough that results can be checked against the standard-candle expectation. The validation applies the full chain blind (template synthesis → mode selection, with Mode S carrying the sparse tail) and reports the Hubble diagram against ΛCDM, the residual rms by redshift bin and by mode, the Hsiao-versus-SALT3-NIR template systematic, and the failure census with named gates. Passing criteria are honest rather than cosmetic: residual rms consistent with the low-z dispersion of §6 plus the propagated sparse-mode errors, and no mode- or z-dependent bias exceeding its error.

Application scope. With filters registered, the identical call serves Rubin (ugrizy to $z \sim 0.4$ in rest-B–V; deeper with y), Roman WFI (rest-frame B–V to $z \gtrsim 2$), and JWST NIRCam follow-up photometry. The cookbook's rule of thumb: choose observer bands so that 4400(1+z) Å and 5500(1+z) Å each fall *inside* the covered wavelength range — synthesis interpolates well and extrapolates badly, and the pipeline gates (`rest_band_uncovered`) when a rest band would require extrapolating the template beyond the observed filters.

8.1 The iterative template fit, and why B_BV0.6 is stable against template details

The SALT3-NIR parameters cannot be fixed in one pass, because the K-correction and the light-curve fit depend on each other. The engine iterates (PI prescription):

1. make a first guess of (x_1, c) — from the raw observed decline and color;
2. compute the K-correction / rest-frame synthesis with the template at that (x_1, c);
3. fit the light-curve template to the synthesized rest-frame light curves to derive new (x_1, c);
4. return to step 2; exit when (x_1, c) converge (default: $|\Delta x_1| < 0.05$ and $|\Delta c| < 0.01$, typically 2–4 iterations; the pipeline gates with `kcorr_no_convergence` otherwise).

Convergence is benign for a reason worth stating plainly: the K-correction is essentially a color correction. Once the template — *any* reasonable Ia template — is morphed to match the observed colors, the synthesized B magnitude at the fiducial color B−V = 0.6 is a stable quantity, insensitive to the residual details of the template. The fiducial-color evaluation point is what buys this: template imperfections move points *along* the color–magnitude locus far more than they move the locus itself, and reading the locus at fixed color cancels the along-track freedom.

This is not an assumption but a measurable property, and the pipeline measures it: the template-insensitivity test recomputes the synthesis with templates of deliberately different (x_1, c) — and with the Hsiao template in place of SALT3-NIR — each morphed to the same observed colors, and reports the spread of the resulting B_BV0.6 as the template systematic (superseding the simpler Hsiao-versus-SALT3-NIR delta as the quoted systematic; the pass criterion is that this spread be small against the photometric error, which is the quantitative form of the stability claim above).

10.1 Worked example: the SDSS-II validation run (v0.2.0)

Blind run of `examples/sdss_validation.py` on 60 spectroscopically confirmed SDSS-II SNe Ia (z 0.05–0.35; public release of Sako et al. 2018 — see the script header for the exact source URLs), engine `hsiao` primary with the §8.1 insensitivity test per object:

- 36 of 60 yield distances — modes: L ×17, S ×19. Failure census (all named): `no_bracket` ×11 (blue branches never reaching B−V = 0.6 inside the window), `rms_gate` ×9, `window_empty` ×3, `too_few_nights` ×1.
- Hubble residuals about ΛCDM ($H_0 = 72$, $\Omega m = 0.3$): grey zero-point offset −0.17 mag (the transfer of the Vega/W03 calibration through synthesis; documented in KNOWN_ISSUES, not recalibrated away), rms about the median 0.44 mag overall: mode L 0.24 mag, mode S 0.56 mag; by redshift bin 0.33 / 0.53 / 0.38 (0.05–0.15 / 0.15–0.25 / 0.25–0.35).
- Error-budget honesty: χ^2/dof = 3.9 (L) and 9.4 (S) — the propagated errors underestimate the observed scatter by ≈2× (L) and ≈3× (S). The L-mode excess over the §6 calibration dispersion plausibly carries SMP photometry systematics and the host-extinction estimator applied to synthesized peak magnitudes; the S-mode excess concentrates at high color leverage (the §9 warning realized: single points far from the fiducial color). The §10 passing criterion is NOT met at v0.2.0; the numbers are reported as measured.
- Template-insensitivity (§8.1): median B_BV0.6 spread over {$x_1 \pm 0.5$, c ± 0.05, Hsiao} = 0.056 mag, 90th percentile 0.091 — the stability claim holds at the few-hundredths level, and per-object spreads are carried as `K_systematic`.

SALT3 comparison (same 60-SN subset; examples/sdss_salt3_comparison.py) SALT3 (sncosmo; Tripp with alpha = 0.14, beta = 3.1, M_B = −19.36; validity cuts $|x_1| < 3$, $|c| < 0.3$) fits 56 of 60; the overlap with CMAGIC's 36 is 34 objects. On the overlap, each method's own grey offset removed:

method	N	grey offset	rms	χ^2/dof	rms (mode-L subset)
SALT3	34	+0.34	0.142	11.2*	0.093
CMAGIC	34	−0.20	0.315	4.3	0.241

*SALT3's χ^2/dof uses its formal fit errors with no intrinsic-scatter term; adding the usual σ_int ≈ 0.1 brings it to ≈1.3.

Residual correlation: Pearson r = −0.01 ± 0.18 — consistent with zero, formally even below the ~0.15 of W03/WS06 at low z, though CMAGIC's larger independent noise dilutes r at this depth: the data are consistent with the two methods sharing little beyond the photometry. Five of 34 objects show $|\Delta\mu| > 3\sigma$ (all with CMAGIC fainter than SALT3 at low z, K_systematic ≤ 0.08 — so not template-choice); and the $\Delta\mu$ panel exposes a mode-dependent relative offset: mean μ_CMAGIC − μ_SALT3 = −0.37 (mode L) versus −0.71 (mode S), an internal L-versus-S inconsistency of ≈ 0.34 mag that is invisible in CMAGIC's own Hubble scatter but obvious against the common SALT3 reference.

Honest paragraph. On sparse, cross-filter SDSS-II data at v0.2.0, CMAGIC is not competitive with SALT3 as a distance estimator: 2–3× the residual rms on the same objects, a lower yield (36 vs 56), and an unresolved L-versus-S zero-point split — SALT3 was trained end-to-end for exactly this regime, while the CMAGIC chain here transfers a low-z, Vega-calibrated, two-band construction through template synthesis. What the comparison buys is what W03/WS06 originally claimed and the r ≈ 0 confirms: the two estimators are statistically independent, so CMAGIC remains valuable as a cross-check and systematics probe (its per-object K_systematic and named gates localize what SALT3 absorbs silently), and the identified defects — the sparse-mode error model, the synthesized-peak extinction estimator, and the mode-dependent zero point — are concrete v0.3 targets rather than fundamental limits.

v0.3 update to the SALT3 comparison With the boundary-hit gate (6 stretch-at-bound objects removed, including two of the former >3σ outliers) and the §6.1 fitted corrections applied, the CMAGIC sample tightens from rms 0.32 (v0.2 overlap) to 0.21 mag overall (0.165 mode L), the L-versus-S zero-point split is absorbed into delta_S = −0.10 ± 0.06, and the per-mode error budgets close (chi^2/dof = 1 by construction of the inflation). SALT3 remains tighter (0.14/0.09); the gap after correction is ≈1.5× rather than ≈2–3×.

10.2 Scatter-statistics convention

Two scatter statistics are quoted throughout, and every table states which it uses. The unweighted rms is the plain standard deviation of the Hubble residuals; the error-weighted rms is $(\Sigma w\, r^2/\Sigma w)^{1/2}$ with $w = 1/\sigma^2$ and each method's *weighted* grey offset removed. Fits (the sample standardization of §6.1, and every regression in the pipeline) are always error-weighted; scatter tables in earlier sections quoted the unweighted rms unless marked. The convention follows Wang, Strovink et al. (2006): their §2.3 χ^2-rescaling of fit errors is the per-fit $\sqrt{(\chi^2/\mathrm{dof})}$ normalization of §9.1 here, and their intrinsic-noise terms (added until χ^2/dof ≈ 1) are the sample-level inflation of §6.1. WS06's Hubble-flow variance also carries a peculiar-velocity term (5/ln 10)·v_pec/(cz) with v_pec fitted (their 95% upper limit 486 km s^{-1}); at this subsample's z ≥ 0.06 a 300 km s^{-1} term contributes only 0.006–0.035 mag against per-object errors of 0.15–0.27 mag, so it is omitted here — include it for any sample reaching below z ≈ 0.03. On the v0.3 standardized SDSS-II sample, both are shown in the Hubble figure: SALT3 0.143 unweighted / 0.133

weighted (n = 56); CMAGIC mode L 0.165 / 0.170 (n = 16); mode S 0.263 / 0.272 (n = 12). With the fitted corrections and honest error inflation, mode L is within ~30% of SALT3's scatter on the same photometry; the sparse mode remains information-limited, as its error bars now correctly reflect.

v0.3.1 note to the SALT3 comparison With the per-fit error normalization and the boundary gates, the overlap is 28 objects; CMAGIC rms 0.274 (χ^2/dof 3.25 pre-standardization; grey −0.07), residual correlation with SALT3 r = +0.01 ± 0.20, three >3σ Δμ outliers. Every figure now carries per-point error bars (x and y), including all 60 per-object panels (failed objects get a named failure panel), with χ^2/dof and the applied error scale printed in each annotation.

10.3 Weighted Hubble residuals (v0.3.1)

Inverse-variance-weighted mean residuals against ΛCDM ($H_0 = 72$, Ω_m = 0.3), errors of the mean scaled by $\sqrt{(\chi^2/\mathrm{dof})}$ where above 1. Standardized CMAGIC (M_0 removed by the fit): all −0.001 ± 0.038 (χ^2/dof 0.98, n = 28); mode L −0.000 ± 0.043 (1.00, 16); mode S −0.003 ± 0.083 (1.03, 12) — the per-mode χ^2/dof ≈ 1 is an outcome of the v0.3.1 error model, not a construction. By redshift bin: +0.036 ± 0.053 (z 0.05–0.15), −0.061 ± 0.071 (0.15–0.25), +0.073 ± 0.140 (0.25–0.35): no redshift trend, and z was never a fitted covariate.

The SALT3 redshift drift, diagnosed. On the same photometry SALT3's weighted residual with *fixed* fiducial Tripp coefficients (α = 0.14, β = 3.1) drifts by +0.161 ± 0.049 (3.3σ) from the low to the high bin (+0.180 ± 0.061 on the 28-object common subset). Most of this is the comparison's own construction, not SALT3, and none of it is a cosmology-grade statement:

- The subsample's covariates drift with redshift exactly as magnitude-limited selection predicts: ⟨x1⟩ = +0.21 → +1.11 and ⟨c⟩ = +0.019 → −0.048 from the low to the high tercile (r(x1, z) = +0.43). Any mismatch between the fiducial coefficients and the ones this subsample prefers therefore maps covariate drift directly into a residual-vs-z slope.
- Refitting (M0, α, β) on the same 28 objects — the identical treatment CMAGIC receives in §6.1 — gives α = 0.097 ± 0.021, β = 2.41 ± 0.20 and cuts the drift to +0.098 ± 0.050 (2.0σ); the residual-vs-z slope falls from +1.03 ± 0.36 to +0.39 ± 0.29 mag per unit z. The coefficient-mismatch × covariate-drift product (+0.085 mag) accounts for the removed part.
- The symmetric comparison is then SALT3 +0.098 ± 0.050 vs CMAGIC +0.016 ± 0.103: a difference of 0.08 ± 0.11 — no significant method discrimination. CMAGIC's flatness currently has ±0.10 mag resolution and cannot claim immunity.
- The residual 2σ drift is what an uncorrected magnitude-limited subsample can show: no simulation-based selection/bias corrections (BBC-style) were applied here, as they are in every published SALT cosmology analysis, and the fits run on a magnitude-space conversion of a cached 60-object subset. Nothing in this table measures cosmology.
- The physics behind the mechanism — why a single fitted β cannot be universal when the dust-to-intrinsic color mixture drifts with redshift — is §5.1's account of the two extinction philosophies.

What survives as the monitoring target: CMAGIC's redshift stability at its present ±0.10 mag resolution, to be retested on a sample with per-bin errors below 0.05 mag. Reproduce the decomposition with `examples/sdss_salt3_drift_diagnostic.py`; the symmetric-treatment Hubble diagram (both methods' standardization fit on the same 28 objects, with the tercile-drift panel) is `examples/make_hubble_residual_fig_symmetric.py` → `figures/sdss_hubble_residuals_symmetric.png`. Under that treatment the weighted rms is 0.080 (SALT3 refit) vs 0.196 (CMAGIC) — both are post-fit quantities (3 and 4 parameters respectively fit on the same 28 objects), so the ~2.4× scatter gap is the honest like-for-like number at v0.3.1. (What that gap measures — and where it goes — is §10.4.)

10.4 DES-SN5YR validation: the scatter gap closes at depth

Data. The public DES-SN5YR release (DES Collaboration 2024; light curves Sánchez et al. 2024; github.com/des-science/DES-SN5YR). The repository caches 122 cosmology-sample SNe Ia at zHEL = 0.05–0.65 — the range where rest-frame B and V are synthesizable from griz — stratified in redshift (all $z < 0.2$, 25 per bin above; `examples/data/des/`), SNANA SMP fluxes converted to magnitudes at SNR ≥ 3, the same convention as the SDSS-II cache so both runs feed the identical pipeline. The release's own SALT3 (x1, c) and bias-corrected MU ride along for cross-checks. Precedent for CMAGIC at these redshifts: Conley et al. (2006) — see Part II's introduction.

Yield. Blind, the chain fits 47/122 (21 L, 26 S). Assisted with release metadata (`t_bmax` = PKMJD; external Δm15 synthesized from the release SALT3 x1, c): 56/122 (23 L, 33 S). SALT3 fits 112/122 of the same tables. Failure census (assisted): dm15_provenance 30, rms_gate 16, no_bracket 13, window_empty 4, too_few_nights 3. An architectural finding: the surviving dm15_provenance failures fire *inside the synthesis stage*, which external metadata cannot reach — passing priors into the synthesis is a v0.4 item. At DES depth CMAGIC's real cost is yield, not scatter.

Contamination. The DES sample is photometrically classified, and one of the 53 common objects is a certain non-Ia by the release's own BEAMS classifier (snid 1770639, P(CC) = 1.00; published residual +1.07 mag). It is an outlier in *both* methods' panels, but it damages the two weighted-rms figures very differently: SALT3's tight formal error makes the outlier expensive, while CMAGIC's inflated mode-S error had already downweighted it — a first parity-looking table built without the classification was an artifact of exactly this. Objects with P(CC) > 0.5 are now excluded (`data/des/des_probcc.csv`); one borderline object (P(CC) = 0.12) is kept.

Symmetric comparison ($n = 52$ clean common objects). SALT3 Tripp coefficients refit on the sample ($\alpha = +0.095 \pm 0.025$, $\beta = +2.88 \pm 0.24$, $\sigma_{int} = 0.152$); CMAGIC standardized per §6.1 ($\alpha_C = +0.063 \pm 0.021$, $\beta_C = +0.63 \pm 0.21$, $\delta_S = -0.044 \pm 0.040$; inflation L 1.45, S 1.44; post-fit r(x1) = r(c) = 0). Weighted rms:

estimator	weighted rms
SALT3, release-grade (bias-corrected MU, full-flux photometry)	0.151
SALT3, refit on sample (same input tables)	0.158
CMAGIC mode L only ($n = 23$)	0.198
CMAGIC standardized, all	0.218
CMAGIC mode S only ($n = 29$)	0.235

The clean gap is 1.25× for mode L and 1.4× overall — roughly half the SDSS 2.4×, but not parity.

Published scatter, for reference (computed from the release's own MURES over the full sample): DES SNe, plain rms 0.276 unweighted / 0.221 weighted — a number that *includes* the core-collapse contamination the cosmology likelihood downweights rather than removes; weighting by the release's own BEAMS P(Ia), 0.15 mag. External low-z subsample: 0.116. Per-z-bin core rms (|residual| < 0.5): 0.145–0.178. Our clean-set numbers sit at this published per-object level.

Redshift drift. With the contaminant removed, the §10.3 pattern reproduces on DES: fixed-fiducial SALT3 drifts $+0.168 \pm 0.059$ (2.9σ) low→high tercile, the sample refit halves it ($+0.105 \pm 0.057$, 1.9σ), CMAGIC is flat (+0.032 ± 0.074), and the release's bias-corrected MU is flat (+0.030) — the survey's simulation-based corrections remove for SALT3 a drift that CMAGIC does not exhibit in the first place. (The low-z contaminant had masked this in the 53-object table, which read flat everywhere.)

The resolution of the 'mystery' (CMAGIC beats peak methods at low z in W03/WS06, loses at 2.4× on SDSS): the gap tracks the signal-to-noise contrast between the light-curve peak and the CMAGIC window, which sits 1.5–2 mag below it. Direct evidence: the median fit error on B_BV0.6 rises 0.055 → 0.074 → 0.106 → 0.148 mag across the DES z bins (photon-noise floor), and SALT3's fitted σ_int jumps from 0.064 (SDSS subset) to ≈ 0.15 (DES subset) — on DES everyone is photometry-limited and the gap halves. At low z both regions have high SNR and CMAGIC's smaller intrinsic dispersion wins (W03); at intermediate z on shallow imaging SALT3's peak-anchored, all-epoch fit wins most strongly (SDSS, §10.3); at depth the field partially levels (DES), with the remaining 1.25–1.4× owed to the window's photon-noise floor and the sparse mode. The corollary is the independence budget: the CMAGIC–SALT3 residual correlation is r = +0.01 ± 0.20 on SDSS but r = +0.67 ± 0.08 on DES — shared photon noise dominates at depth, so CMAGIC's value as an *independent* cross-check is a property of the intrinsic-limited regime (low z, or deep imaging of bright targets), while at the survey limit it functions as a consistency check with different systematics, not an independent one.

Cross-check of our quick SALT3 fits against the release's: Δc median −0.001 (rms 0.068), Δμ rms 0.156 about a +0.31 grey offset (their bias corrections and zero-point convention). Reproduce: `examples/des_validation.py [--assisted]`, `examples/des_salt3_comparison.py`, `examples/des_standardize_and_fig.py` → `figures/des_hubble_residuals_symmetric.png`.